

MATH8009: Matrix Operations - Problem Sheet 8

Q1. Determine

$$(a) A + B, \quad (b) 2A - 3B, \quad (c) AB$$

for each pair below:

$$i) \quad A = \begin{pmatrix} 6 & -1 \\ 2 & 4 \\ -3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ -1 & 5 \\ 1 & 10 \end{pmatrix}$$

$$ii) \quad A = \begin{pmatrix} 5 & 1 \\ -3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & -2 \\ 2 & 0 \end{pmatrix}$$

$$iii) \quad A = \begin{pmatrix} 5 \\ -3 \\ 6 \end{pmatrix}, \quad B = \begin{pmatrix} 2 \\ 5 \\ -4 \end{pmatrix}$$

Q2. Find (a) c_{21} and (b) c_{13} where $C = 2A - 4B$, with

$$A = \begin{pmatrix} 5 & 4 & 3 \\ -2 & 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 & 7 \\ 6 & 4 & 2 \end{pmatrix}$$

Q3. Find (a) c_{21} , (b) c_{11} , (c) d_{22} and (d) d_{12} where $C = 5A + 2B$ and $D = BA$, with

$$A = \begin{pmatrix} 4 & 11 \\ 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ -4 & 6 \end{pmatrix}$$

Q4. Let A and B be 4×5 matrices, C be a 5×2 matrix, D be a 4×2 matrix and E be a 5×4 matrix. Determine which of the following matrix expressions are defined. For those which are defined, give the size of the resulting matrix.

$$(i) BA, \quad (ii) AC + D, \quad (iii) AE + B, \quad (iv) AB + B, \quad (v) EAC, \quad (vi) (A^T + E)$$

Q5. Solve for x, y and z in the matrix equation

$$4 \begin{pmatrix} x & y \\ z & -1 \end{pmatrix} = 2 \begin{pmatrix} y & z \\ -x & 1 \end{pmatrix} + 2 \begin{pmatrix} 4 & x \\ 5 & -x \end{pmatrix}.$$

MATH8009: Matrix Operations - Problem Sheet 8

Q6. Solve for x, y, z and w in the matrix equation

$$\begin{pmatrix} w & x \\ y & x \end{pmatrix} = \begin{pmatrix} -4 & 3 \\ 2 & -1 \end{pmatrix} + 2 \begin{pmatrix} y & w \\ z & x \end{pmatrix}.$$

Q7. Show that $B = A^{-1}$ by checking the products AB and BA and verifying that both yield the identity matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Q8. Find the inverse of the following matrices using the inverse matrix formula

a) $\begin{pmatrix} 5 & 3 \\ 0 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 2 & -4 \\ 1 & 2 \end{pmatrix}$ c) $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$

Q9. Find the inverse of the following matrices using the Gauss-Jordan Method

a) $\begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 5 & 3 & 3 \\ 2 & 3 & 8 \\ 1 & 1 & 2 \end{pmatrix}$ c) $\begin{pmatrix} 0 & 0 & 1 \\ 2 & -1 & 3 \\ 1 & 1 & 4 \end{pmatrix}$